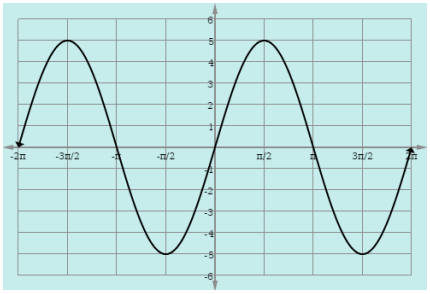
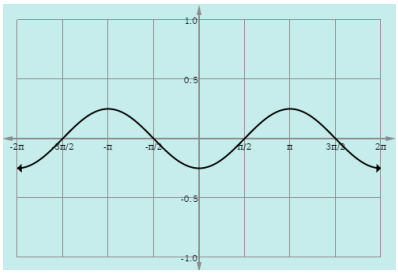
Practice

Find the amplitude, domain, range, y-intercept, x-intercepts, maximums, and minimums of each function. Then, write an equation of the function.

1)

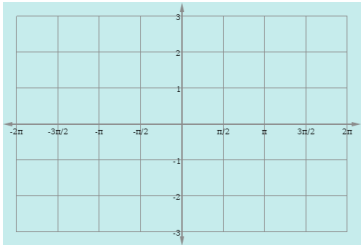


2)

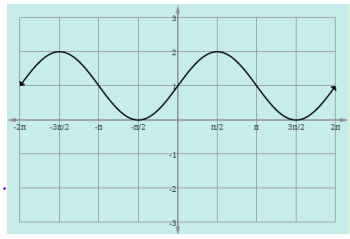


3) Find the amplitude of the function and sketch its graph. Then, find the domain, range, y-intercept, x-intercepts, maximums, and minimums.

$y = -2.5 \sin(x)$



4) Observe the function below:

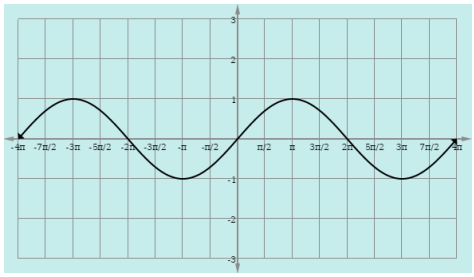


a) What is the range of the function?

b) What is the amplitude of the function?

c) What might be the equation of the function?

5) Observe the function below:



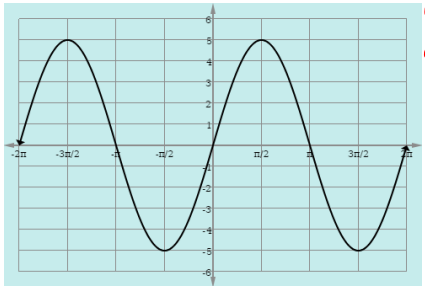
a) Write an expression that represents all of the graph's x-intercepts.

b) Write an expression that represents all of the graph's maximums.

Practice

Find the amplitude, domain, range, y-intercept, x-intercepts, maximums, and minimums of each function. Then, write an equation of the function.

1)



amp: 5 dom: $\mathbb{R}$

range: $[-5,5]$

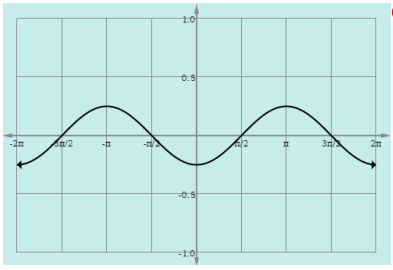
y-int: $(0,0)$

x-ints: $k\pi$

max: $\frac{\pi}{2} + 2k\pi$

min: $\frac{3\pi}{2} + 2k\pi$

2)



amp: 0.25 dom: $\mathbb{R}$

range: $[-0.25, 0.25]$

y-int: $(0, -0.25)$

x-ints: $\frac{\pi}{2} + k\pi$

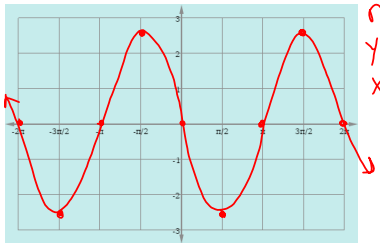
max: $\pi + 2k\pi$

min: $2k\pi$

$y = -0.25\cos(x)$

3) Find the amplitude of the function and sketch its graph. Then, find the domain, range, y-intercept, x-intercepts, maximums, and minimums.

$y = -2.5\sin(x)$



amp: 2.5 dom: $\mathbb{R}$

range: $[-2.5, 2.5]$

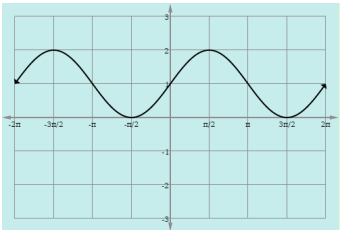
y-int: $(0,0)$

x-int: $k\pi$

max: $\frac{3\pi}{2} + 2k\pi$

min: $\frac{\pi}{2} + 2k\pi$

4) Observe the function below:

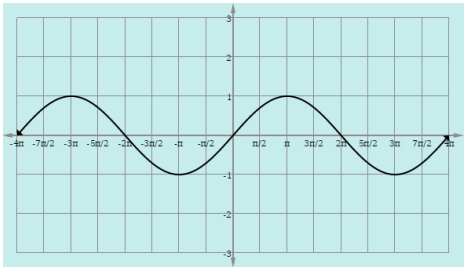


a) What is the range of the function? $[0, 2]$

b) What is the amplitude of the function? 1

c) What might be the equation of the function? $y = \sin(x) + 1$

5) Observe the function below:



a) Write an expression that represents all of the graph's x-intercepts.

$2k\pi$

b) Write an expression that represents all of the graph's maximums.

$\pi + 4k\pi$